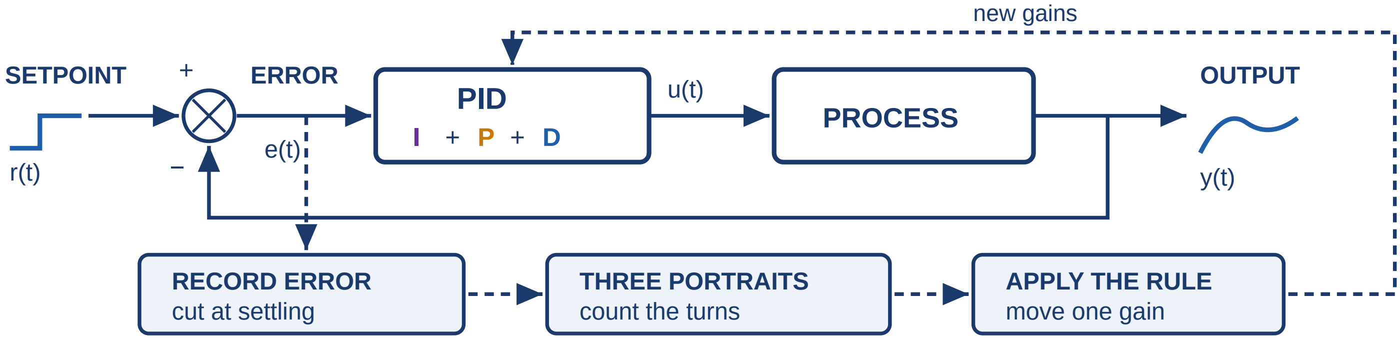
SPIN TUNING

STEP-RESPONSE PHASE-PORTRAIT INSPECTION

Tune a PID loop from one step response. No model, no relay test, no optimizer.



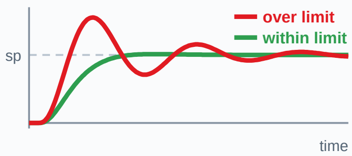
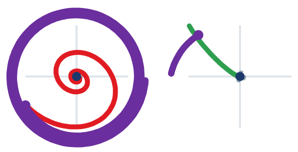
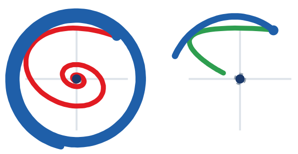
$$u(t) = K_i \int e(\tau)d\tau + K_p e(t) + K_d de(t)/dt$$

each gain multiplies one signal — and each portrait plots that same signal

K_i Integral → Γ_0
K_p Proportional → Γ_1
K_d Derivative → Γ_2

WHAT EACH PORTRAIT SHOWS

K_i K_p K_d — the band, used for the outer angle arc. **red** curve = over the limit, cut that gain. **green** curve = within limit, safe to raise.

PORTRAIT	STEP RESPONSE	PACHNER PLOTS — OVER / WITHIN	READING
Γ_0 K_i band ∫e vs e	 K = (0.35, 0.05, 0.5) K = (0.2, 0.02, 0.7) Both portraits come from these two runs.	 N_0 = 2.41 N_0 = 0.12	✔ Slow cycling — the loop hunts around setpoint ✔ Limit $\bar{N}_0$ **= 0.50** = 2 quarter turns ✔ Over limit: cut K_i ↓, raise bands below ↑
Γ_1 K_p band e vs Δe		 N_1 = 2.54 N_1 = 0.1	✔ Ringing — decaying oscillation after the step ✔ Limit $\bar{N}_1$ **= 0.75** = 3 quarter turns ✔ Over limit: cut K_p ↓, raise bands below ↑
Γ_2 K_d band Δe vs Δ²e		 N_2 = 2.58 N_2 = 0.29	✔ Buzzing — fast chatter on the transient ✔ Limit $\bar{N}_2$ **= 1.00** = 4 quarter turns ✔ Over limit: cut K_d ↓, raise bands below ↑

TUNING GUIDE (SPIN ITERATION)

1 Step it
Close the loop with any gains, step the setpoint, record **e**

2 Trim it
Screen for runaway, cut where |e| last leaves **2%** of peak

3 Count it
Draw the three portraits, count turns N_0 N_1 N_2

4 Move it
Use the table below to move K_i K_p K_d, then repeat

RULE — FIRST MATCHING ROW WINS	K_I	K_P	K_D	READING
record grows	⇓	⇓	⇓	runaway — halve all
Γ_0 loops	↓	·	·	slow cycling — less reset
Γ_1 loops	↑	↓	·	ringing — less gain
Γ_2 loops	↑	↑	↓	buzzing — less rate
none loops	↑	↑	↑	all quiet — tighten

TIPS
- ✔ Cut only the **lowest** objecting band — higher ones may be leakage
- ✔ Bands below the fault were just measured safe, so raise them
- ✔ Counts carry no units: the same limits fit every loop
- ✔ Any start works: the screen halves gains until the record settles
- ✔ No stopping test: stop at any iteration

↑ ×γ ↓ ÷γ ⇓ halve · hold | γ = 1/(1−β), β = 0.1; $\bar{N}$ = (0.5, 0.75, 1.0), ε = 0.1, δ = 0.02 | solid = counted arc, grey dash = discarded tail, outer arc = the angle actually swept

Pachner, Otta, Dostál, Havlena — "Model-Free PID Tuning by Step-Response Inspection," submitted to *Journal of Process Control*, 2026. · CTU Prague ·
robopid.uceeb.cvut.cz